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SOIL Research Methodology White Paper

A Framework for Systematic Study of Organizational Mortality

Social Organizational Intelligence Lab

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Abstract

This white paper presents the methodological foundation for SOIL (Social Organizational Intelligence Lab), a research initiative establishing the systematic study of organizational mortality as a scientific discipline. We propose a framework-agnostic approach to collecting, analyzing, and interpreting organizational autopsy data at scale. The methodology addresses fundamental challenges in organizational research: survivor bias, self-report validity, temporal reconstruction accuracy, and multi-stakeholder perspective integration. By combining structured data collection with therapeutic interview techniques, computational text analysis, and multi-framework analytical approaches, SOIL aims to build the empirical foundation for what we term “Organizational Medicine” — the systematic understanding, prediction, and prevention of organizational death.

Keywords: organizational mortality, failure analysis, autopsy methodology, organizational health, research design, longitudinal data collection

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1. Introduction and Rationale

1.1 The Knowledge Gap

Organizational mortality represents one of the most consequential yet understudied phenomena in social science. While estimates suggest that 90% of startups fail and organizational death affects millions of individuals annually, our systematic understanding of why organizations die remains remarkably primitive.

Current knowledge suffers from several critical deficiencies:

Survivor Bias Dominance. The vast majority of organizational research examines successful organizations, extrapolating “success factors” from survivors. This approach fundamentally cannot distinguish between factors that contribute to success and factors that are merely common among both survivors and casualties.

Anecdotal Evidence Base. Existing failure knowledge consists primarily of post-hoc narratives, case studies selected for dramatic value, and self-serving founder accounts. These sources lack systematic structure, standardized measurement, and independent verification.

Temporal Compression. When failures are studied, analysis typically focuses on proximate causes — the final crisis that precipitated closure. The developmental trajectory leading to vulnerability remains largely unexplored.

Single-Perspective Limitation. Most failure accounts represent a single stakeholder view (typically the founder), missing the multi-dimensional reality of organizational dysfunction.

1.2 The Medical Analogy

Modern medicine developed through systematic autopsy — the careful examination of deceased bodies to understand disease processes. Before autopsy became standard practice, medicine relied on theory, speculation, and case reports. The shift to systematic post-mortem examination created the empirical foundation for pathology, which in turn enabled diagnosis, prognosis, and treatment.

Organizational science currently resembles pre-autopsy medicine. We possess abundant theory about organizational health but lack the systematic empirical foundation that would allow us to:

- Classify organizational pathologies taxonomically
- Identify early warning indicators with predictive validity
- Distinguish survivable crises from terminal conditions
- Develop evidence-based intervention protocols

SOIL proposes to address this gap by establishing organizational autopsy as a rigorous research practice.

1.3 Research Objectives

Primary Objective: Build a comprehensive, verified, multi-framework database of organizational mortality cases sufficient to enable pattern identification, hypothesis testing, and predictive model development.

Secondary Objectives:

1. Develop and validate instruments for organizational autopsy data collection
2. Establish reliability standards for retrospective organizational assessment
3. Create taxonomies of organizational pathology
4. Identify early warning indicators with predictive validity
5. Generate testable hypotheses about organizational mortality mechanisms
6. Provide therapeutic value to participants while maintaining research rigor

2. Theoretical Foundations

2.1 Framework Agnosticism as Methodological Principle

A critical methodological decision shapes SOIL's approach: we deliberately avoid commitment to any single theoretical framework for understanding organizations.

Rationale: Existing organizational theory offers numerous competing frameworks — systems theory, institutional theory, population ecology, resource-based view, organizational learning, complexity theory, and many others. Each framework emphasizes different aspects of organizational functioning and would therefore structure data collection differently.

Premature commitment to any single framework risks:

- **Confirmation bias:** Collecting data that supports the chosen framework while missing contradictory evidence
- **Construct validity threats:** Imposing categories that may not reflect organizational reality
- **Analytical lock-in:** Preventing comparison of explanatory power across frameworks
- **Field fragmentation:** Contributing to theoretical silos rather than cumulative knowledge

Our Solution: Collect data in relatively neutral formats that can subsequently be analyzed through multiple theoretical lenses. This approach:

- Preserves information that any single framework might discard
- Enables post-hoc comparison of framework explanatory power
- Allows patterns to emerge from data rather than being imposed
- Builds credibility through methodological transparency

2.2 Multi-Framework Analytical Strategy

Rather than choosing a framework, we apply multiple frameworks as analytical tools to the same dataset:

Biological/Systems Framework Treats organizations as complex adaptive systems with interdependent subsystems (analogous to organ systems). Analyzes failure as system breakdown, cascade effects, and homeostatic failure.

Traditional Business Functions Framework Examines organizations through conventional functional areas: sales, marketing, operations, finance, HR, etc. Analyzes failure as functional deficiency or coordination breakdown.

Institutional Theory Framework Examines legitimacy, isomorphism, and field-level dynamics. Analyzes failure as legitimacy loss or institutional misfit.

Resource Dependence Framework Examines critical resource flows and dependencies. Analyzes failure as resource starvation or dependency vulnerability.

Population Ecology Framework Examines environmental selection, carrying capacity, and competitive dynamics. Analyzes failure as environmental misfit or niche disappearance.

Organizational Learning Framework Examines knowledge acquisition, error correction, and adaptation. Analyzes failure as learning failure or adaptation deficiency.

Complexity Theory Framework Examines emergence, feedback loops, and non-linear dynamics. Analyzes failure as complexity overload or cascade failure.

Each framework generates different hypotheses, different pattern expectations, and different practical implications. By maintaining framework agnosticism at the data collection stage, we enable rigorous comparison.

2.3 Temporal Dynamics Model

Organizational mortality is not an event but a process. Our theoretical model posits three distinct phases requiring different data collection approaches:

Phase 1: Genesis and Early Development The period from founding through initial operations. Data collection focuses on founder characteristics, founding conditions, initial resource endowments, and early strategic choices.

Phase 2: Peak Operations The period when the organization achieved its highest functionality. This becomes our primary temporal anchor for comparative analysis — capturing organizational structure and health at maximum capability.

Phase 3: Decline and Termination The period from peak through closure. Data collection focuses on degradation dynamics, crisis events, response patterns, and terminal processes.

This tripartite temporal model enables: - Within-case trajectory analysis - Cross-case comparison at standardized temporal points - Distinction between initial conditions, steady-state vulnerabilities, and proximate causes

3. Research Design Philosophy

3.1 Mixed Methods Integration

SOIL employs a convergent mixed-methods design, collecting both quantitative and qualitative data through integrated instruments rather than parallel streams.

Quantitative Components: - Structured organizational metrics (headcount, revenue, funding, etc.) - Standardized function assessment scales - Timeline markers and durations - Likert-scale health indicators

Qualitative Components: - Open-ended narrative responses - Critical incident descriptions - Meaning-making and attribution accounts - Contextual explanations

Integration Strategy: Quantitative data provides the skeleton for systematic comparison; qualitative data provides the flesh for understanding mechanisms. Neither alone is sufficient.

3.2 Retrospective Design Considerations

Organizational autopsy is inherently retrospective — we study organizations after they have died. This creates well-known methodological challenges that we address through specific design features.

Challenge: Memory Decay and Reconstruction

Humans reconstruct memories rather than replay recordings. Over time, memories become simplified, schematized, and influenced by subsequent events and current beliefs.

Mitigations: - Temporal anchoring at Peak Operations (a salient, typically positive memory) - Multiple external memory cues (documents, timelines, contemporaneous records) - Focus on structural/factual data before interpretive accounts - Collection of supporting documentation where available - Multi-stakeholder verification for key facts

Challenge: Hindsight Bias

After knowing an outcome, people overestimate how predictable it was and reconstruct their past beliefs to align with current knowledge.

Mitigations: - Explicit temporal framing (“What did you believe at the time?”) - Separation of contemporaneous assessments from retrospective judgments - Direct questioning about surprises and expectation violations - Collection of contemporaneous documents (pitch decks, plans) for comparison

Challenge: Self-Serving Attribution

Founders may attribute failure to external factors (bad luck, market conditions) while taking credit for any successes.

Mitigations: - Multi-stakeholder verification - Structured prompts for internal and external factors - Pattern analysis across cases (consistent external attributions become suspicious) - Documentary evidence for claimed external events

3.3 Unit of Analysis

The primary unit of analysis is the **organizational mortality episode** — the complete arc from an organization’s founding through its termination, as documented through autopsy.

Secondary units include: - **Functional subsystems** within organizations - **Critical events** within organizational timelines - **Stakeholder perspectives** on the same organization - **Environmental contexts** surrounding multiple organizations

This multi-level analytical capability enables both intensive within-case analysis and extensive cross-case comparison.

3.4 Comparison Strategy: Survivors and Near-Death Cases

Understanding why organizations die requires comparison with organizations that survived similar conditions. SOIL addresses this through a multi-pronged comparison strategy:

Near-Death Survivor Analysis

Organizations that experienced severe crises but survived provide critical counterfactual evidence. We will collect data from:

- Organizations that survived conditions similar to those that killed other organizations
- Organizations that recovered from near-terminal states
- Organizations that successfully navigated specific crisis types

Data Collection Approach:

Source	Method	Purpose
Retrospective survivor interviews	Same instrument modules, adapted framing	Direct comparison on same dimensions
Longitudinal monitoring	Ongoing health assessment of at-risk organizations	Prospective validation of warning indicators
External datasets	Business registries, accelerator cohort data	Population-level survival rate baselines

Matched Comparison Design:

Where possible, we will construct matched pairs or groups: - Same industry, geography, and founding period - Similar initial conditions (funding, team size) - Different outcomes (death vs. survival)

This enables more rigorous isolation of mortality-associated factors.

Open Questions for Collaborative Development:

The optimal design for survivor comparison studies requires careful methodological consideration. We invite collaboration from researchers with expertise in: - Matched-pair and case-control designs in organizational research - Counterfactual analysis methods - Prospective cohort study design - Survival analysis in business contexts

Specific questions include: What matching criteria best balance precision and feasibility? How do we handle the selection bias inherent in survivor willingness to participate? What prospective data collection is feasible given resource constraints?

4. Data Collection Methodology

4.1 Sampling Strategy

Target Population: Organizations that have permanently ceased operations after a period of active functioning.

Inclusion Criteria: - At minimum: 2+ people involved at some point (co-founder, employee, contractor) OR external stakeholders (paying customers, investors, suppliers) - Clearly definable founding and termination dates - Identifiable founder(s) or senior leader(s) willing to participate - Sufficient documentation or verification sources

Exclusion Criteria: - Solo side projects without external stakeholders - Organizations still operating (even if transformed) - Mergers and acquisitions (unless the original entity genuinely ceased to exist) - Temporary projects with predetermined endpoints

Sampling Approach: Purposive sampling with stratification by: - Organization type (tech, services, manufacturing, NGO, media, e-commerce) - Geographic region - Funding stage / size at peak - Industry sector - Time since closure

Initial phases prioritize breadth across strata; later phases enable targeted sampling to fill gaps and test specific hypotheses.

Non-Response Analysis Protocol:

To assess and mitigate selection bias, we implement systematic non-response analysis:

1. **Tracking:** Record all recruitment attempts, including non-responses and refusals
2. **Comparison:** Compare available characteristics of non-respondents (from public sources: LinkedIn, Crunchbase, press) with respondents
3. **Refusal reasons:** When possible, collect brief reasons for non-participation
4. **External benchmarking:** Compare sample composition against external data sources (business registries, accelerator databases, industry reports) to assess representativeness
5. **Sensitivity analysis:** Model how results might differ under various assumptions about non-respondent characteristics
6. **Weighting consideration:** Develop propensity weights to adjust for known selection factors where appropriate

Reporting Commitment: All publications will include transparent reporting of: - Response rates by recruitment channel - Known differences between respondents and non-respondents - Limitations on generalizability based on sample composition

4.2 Data Collection Instruments

Data collection occurs through a modular wizard system, with each module capturing distinct aspects of organizational life and death.

Module 1: Founder Context (15-20 minutes) **Purpose:** Capture founder characteristics, personal dynamics, and human cost.

Background: - Prior experience (first-time vs. serial) - Domain expertise level - Life circumstances at founding

Founding Dynamics: - Entry mode (full-time vs. gradual) - Team composition (solo, co-founder, team) - Role clarity

Evolution: - Motivation trajectory - Co-founder relationship dynamics (if applicable) - Personal investment intensity

Personal Cost: - Health impact - Relationship impact - Financial impact - Recovery timeline

Personal Events: Life events affecting founder capacity: Health, Family, Life changes, Other commitments. Each includes impact on capacity and organizational adaptation.

Methodological Notes: - Founder-level data enables analysis of individual vs. organizational factors - Personal cost data addresses an understudied dimension of organizational mortality - Personal events often explain otherwise puzzling organizational dynamics

Module 2: Financial Picture (15-20 minutes) **Purpose:** Capture financial metrics and dynamics from Peak to closure.

Data Structure:

Metrics at Peak (organization-type specific):

Organization Type	Key Metrics
Tech Product	MRR/ARR, burn rate, runway, LTV/CAC, total funding, stage
Services	Monthly revenue, gross margin, utilization, project size, cash cycle
E-commerce	GMV, net revenue, gross margin, CAC, ROAS, inventory turnover
Manufacturing	Revenue, gross margin, COGS%, inventory, working capital, capex
NGO	Annual budget, funding mix, overhead ratio, reserves
Media	Revenue, revenue mix, audience size, CPM/ARPU

Dynamics (Peak → Death): For each key metric: Grew / Stable / Declined / Collapsed

Financial Events: Structured event capture for: Revenue events, Funding events, Cash events, Cost events, Profitability events.

Each event includes: Date, category, specific type, severity (Minor/Significant/Critical), response taken, retrospective assessment.

Document Upload (Optional): P&L statements, bank statements, management dashboards, pitch decks. AI extraction with founder verification.

Document-Report Reconciliation:

When documents are uploaded: 1. AI extracts key metrics from documents 2. System compares extracted values with self-reported values 3. Discrepancies >20% flagged for founder review 4. Founder confirms, corrects, or explains discrepancy 5. Final reconciled values stored with provenance tracking 6. Analysis can stratify by document-verification status

Methodological Notes: - Financial data is the most objective indicator of organizational health - Document upload enables verification of self-reported metrics - Financial events create natural linkage to timeline module

Module 3: Dynamic Picture (20-30 minutes) **Purpose:** Capture the degradation trajectory from Peak to closure.

Pattern-Based Prompting: System analyzes Module 5 (Functional Mapping) health check data to detect cross-functional patterns: - Systemic Turnover (high turnover in 2+ functions) - Widespread Understaffing (understaffed in 2+ functions) - Budget Squeeze (pressure in 2+ functions) - Quality Erosion (issues in 2+ functions) - Leadership Crisis (gaps/vacuum in 2+ functions) - Internal Friction (cross-function conflict in 2+ pairs)

Each detected pattern triggers targeted follow-up questions exploring timing, causation, and response.

Event Timeline: Founder adds significant internal events chronologically, each tagged by: - Category (Team, Product, Operations, Growth Pains, Hostile Actions, Closure) - Specific type (from standardized taxonomy) - Emotional state at time - Retrospective interpretation - Optional narrative details

Methodological Notes: - Pattern-based prompting ensures systematic coverage without exhausting respondents - Event cards provide standardized structure while allowing open-ended elaboration - Emotional state data enables affect analysis - Retrospective interpretation enables comparison of in-the-moment vs. hindsight views

Module 4: Environment Analysis (15-20 minutes) **Purpose:** Capture external conditions and events affecting the organization.

Resource Assessment: For each environmental resource (Customers, Talent, Suppliers, Capital, Infrastructure, Legal/Justice, Regulatory, Tax), assess: - State at Peak (Availability, Cost, Competition) - Change Peak → Death (Improved / Stable / Declined)

External Events: Structured event capture for: Market shifts, Competitive dynamics, Regulatory changes, Capital environment changes, Talent market changes, Macro events, Technology changes, Supplier events, Reputation events, Hostile external actions.

Methodological Notes: - Resource framework adapted from resource dependence theory but collected without theoretical commitment - External events enable environmental context analysis - Separation of internal (Module 3) and external (Module 4) events enables attribution analysis

Module 5: Functional Mapping (30-40 minutes) **Purpose:** Capture comprehensive organizational structure and health at Peak Operations.

Temporal Anchor: Peak Operations

The moment when the organization was operating at its best — maximum functionality before decline began.

Peak Identification Protocol:

To ensure consistent and valid identification of Peak Operations across cases:

1. **Objective markers:** System prompts for key metrics (max headcount, max revenue month, max customer count) to triangulate Peak timing
2. **Founder identification:** Founder explicitly identifies when they felt the organization was “working best”
3. **Consistency check:** If objective markers and subjective identification diverge significantly (>6 months), system flags for clarification
4. **Ambiguous cases:** When multiple peaks exist or decline was gradual:
 - Founder selects primary peak with explanation
 - Secondary peak data collected optionally
 - Ambiguity documented for analytical consideration

Data Structure per Function:

Dimension	Measurement	Options/Scale
Execution Model	Categorical	In-house / Outsourced / Hybrid / None
Headcount	Numeric	Count
FTE Allocation	Percentage	Per-person FTE% summed
Formalization	Ordinal	None / Informal / Documented / Tooled / Automated
Ownership	Categorical	Dedicated / Shared / Founder / Nobody
Temporal Span	Stage-based	When started, when stopped
Satisfaction	5-point scale	1 (poor) to 5 (excellent)
Issue Description	Open text	If satisfaction ≤ 3

Health Check Indicators per Function:

Indicator	Measurement	Options
Turnover	Categorical	Low / Normal / High
Staffing	Categorical	Adequate / Understaffed
Budget Pressure	Categorical	None / Some / Severe
Quality Issues	Categorical	None / Some / Serious
Leadership	Categorical	Stable / Gaps / Vacuum
Cross-function Conflict	Binary	Yes / No

Function Taxonomy and Adaptive Display:

The complete taxonomy includes 60+ distinct functions organized hierarchically by domain (Sales, Marketing, Product, Engineering, Operations, Finance, People/HR, Customer Success, Legal, IT/Security) with 3-6 sub-functions per domain.

Adaptive filtering: - Organization type determines base function set (e.g., NGOs see “Fundraising” not “Sales”) - Lifecycle stage affects expected maturity (seed-stage orgs not asked about “Investor Relations”) - Branching logic: if parent function marked “None,” children are skipped - Typical respondent sees 25-35 functions (not all 60+)

Fatigue mitigation: - Progress indicator shows completion percentage - Optional save-and-continue at any point - Estimated time remaining displayed - Functions grouped by domain with domain-level completion feedback

Methodological Notes: - Peak Operations anchor chosen because: (a) it's typically a positive, salient memory; (b) it represents maximum organizational capability; (c) it provides a standardized comparison point across cases - Health Check indicators designed to detect systemic patterns (multiple functions showing same issues) - Function-level granularity enables both aggregated and detailed analysis

Module 6: Narrative (20-30 minutes) Purpose: Capture founder interpretation, lessons, and meaning-making.

Core Questions: - Primary cause of death attribution - Point of no return identification - Preventability assessment - Hindsight learnings (what would do differently, what would do same) - Unexpected discoveries - Advice for future founders - Pride points - Desired legacy

Format Options: - Text responses (self-service) - AI-guided interview (conversational) - Human Pathologist interview (professional support option)

The Human Pathologist interview is a separate service for founders who desire professional support during the reflection process. It is not positioned as a “better” data collection method — all formats yield valid research data.

Methodological Notes: - Narrative module comes last to avoid priming earlier factual responses - Open-ended format captures aspects not anticipated by structured instruments - Multiple format options accommodate different participant preferences and needs

4.3 Verification System

Verification addresses a fundamental challenge in self-report research: respondents may intentionally or unintentionally provide inaccurate information.

Verification Requirement: Each cenotaph requires 3 confirmations from verifiers before publication.

Verifier Categories: - Former employees - Former co-founders - Former customers - Former suppliers - Former investors - Former partners

Verification Process:

1. Founder provides 5-10 contact emails with relationship type
2. Platform sends automated verification requests
3. Verifiers confirm relationship and basic facts (organization name, dates, general nature)
4. Verifiers optionally provide their own perspective on organizational dynamics
5. Discrepancies flagged for review

Multi-Perspective Enhancement: When multiple stakeholders participate fully (not just verification), their accounts are triangulated. Convergent accounts

strengthen confidence; divergent accounts are flagged and explored.

Document Verification: Where available, contemporaneous documents (incorporation papers, dissolution documents, pitch decks, press coverage) provide objective verification of claimed facts.

4.4 Data Quality Tiers

Not all cases achieve equal verification depth. We maintain transparent quality tiers:

Tier	Requirements	Research Use
Bronze	Founder self-report only	Pattern exploration, hypothesis generation
Silver	3+ external verifications	Standard analysis, aggregation
Gold	Verified + documents	High-confidence claims, validation studies
Platinum	Multi-stakeholder full participation	Process reconstruction, deep case study

Research publications specify which quality tiers were included in each analysis.

5. Validity and Reliability Framework

5.1 Construct Validity

Challenge: Do our instruments measure what they purport to measure?

Approaches:

Content Validity: - Instrument development informed by multiple organizational theory traditions - Expert review by organizational scholars, practitioners, and founders - Iterative refinement based on pilot testing - Comprehensive coverage across recognized organizational domains

Convergent Validity: - Multiple indicators for key constructs (e.g., organizational health assessed via function satisfaction, health check indicators, and financial metrics) - Correlation analysis across indicators expected to measure same construct - Multi-stakeholder agreement on factual claims

Discriminant Validity: - Distinct constructs measured by distinct instruments - Factor analysis to confirm dimensional structure - Examination of unexpected correlations for construct contamination

5.2 Internal Validity

Challenge: Can we validly draw causal inferences from retrospective data?

Approaches:

Temporal Ordering: - Explicit temporal markers throughout data collection - Event sequencing within and across modules - Documentary evidence of timing where available

Alternative Explanation Consideration: - Systematic collection of environmental factors - Multi-stakeholder perspectives on causation - Pattern matching across cases (consistent cause-effect patterns across contexts strengthen causal inference)

Dose-Response Relationships: - Examination of whether more severe dysfunction associates with faster/more certain mortality - Analysis of near-death experiences (organizations that survived similar challenges)

Limitations Acknowledged: - Retrospective design fundamentally limits causal inference - We frame findings as patterns and associations requiring prospective validation - Strong causal claims require prospective follow-up studies

5.3 External Validity

Challenge: Do findings generalize beyond our sample?

Approaches:

Stratified Sampling: - Deliberate inclusion across organization types, regions, sizes, and sectors - Subgroup analysis to examine pattern consistency - Transparent reporting of sample composition

Boundary Condition Analysis: - Explicit examination of where patterns do and do not hold - Moderator analysis (e.g., “this pattern appears only in early-stage organizations”) - Cultural and contextual variation examination

Replication: - Multi-region data collection enables cross-cultural replication - Temporal batches enable temporal replication - Transparent methodology enables external replication

5.4 Reliability

Challenge: Would repeated measurement yield consistent results?

Approaches:

Inter-Rater Reliability (IRR):

For qualitative data coding, we employ rigorous IRR protocols:

Component	Specification
Metric	Krippendorff’s alpha (α) for nominal/ordinal data; ICC for continuous
Target threshold	$\alpha \geq 0.70$ for exploratory analyses; $\alpha \geq 0.80$ for confirmatory
Coder training	Standardized training on codebook with practice cases

Component	Specification
Calibration	Initial 10% of cases coded by all coders; IRR calculated
Iteration	If $\alpha < \text{threshold}$, codebook refined, coders re-calibrated
Disagreement resolution	Discussion to consensus; unresolved cases flagged
Drift monitoring	Periodic IRR checks (every 100 cases) to detect coder drift
Reporting	All publications report IRR coefficients for each coded variable

For AI-extracted data: - Human audit of random 5% sample for extraction accuracy - Accuracy threshold: 95% agreement with human extraction - Systematic error analysis when accuracy below threshold

Test-Retest Reliability: - Longitudinal follow-up with subset of participants (6-month, 1-year) - Assessment of account stability over time - Identification of systematically unstable elements

Internal Consistency: - Multi-item scales assessed via Cronbach's alpha - Scale refinement based on reliability analysis - Reporting of reliability coefficients for all scales

Parallel Forms: - AI interview and human interview modalities compared - Assessment of systematic differences by collection method - Method effects modeled in analysis where relevant

6. Analytical Approaches

6.1 Descriptive Analytics

Mortality Demographics: - Distribution of deaths by organization type, region, stage, sector - Temporal patterns (time-to-death distributions, calendar clustering) - Size distributions at founding, peak, and death

Functional Profiles: - Function presence rates by organization type and stage - Health indicator distributions - Formalization and ownership patterns

Financial Patterns: - Typical trajectories by organization type - Crisis event frequency and sequencing - Response pattern effectiveness

6.2 Pattern Recognition

Failure Archetype Identification

Using cluster analysis and latent class methods to identify recurring mortality patterns:

- What combinations of dysfunctions commonly co-occur?
- What temporal sequences characterize different mortality pathways?
- What environmental conditions associate with different mortality types?

Early Warning Signal Detection

Using sequence analysis and survival modeling:

- What observable indicators precede mortality?
- How much lead time do different indicators provide?
- What indicator combinations have predictive validity?

Co-Occurrence and Sequence Analysis

We examine patterns of dysfunction across organizational functions:

- **Co-occurrence patterns:** Which dysfunctions tend to appear together?
- **Temporal sequences:** Which dysfunctions tend to precede others?
- **Spread patterns:** How does dysfunction in one function associate with subsequent dysfunction in related functions?

Methodological caution: Temporal sequence does not establish causation. When dysfunction A precedes dysfunction B, possible explanations include: - A caused B (cascade) - B caused A (reverse causation with reporting lag) - C caused both A and B (common cause) - Coincidence

We report temporal patterns descriptively and avoid causal language unless supported by: - Consistent patterns across diverse contexts - Theoretical plausibility - Corroborating evidence (multi-stakeholder, documentary) - Ideally, prospective validation

6.3 Multi-Framework Comparison

The framework-agnostic data collection enables systematic comparison of theoretical explanatory power:

Procedure: 1. Derive predictions from each framework about expected mortality patterns 2. Operationalize predictions using collected data 3. Assess empirical support for each framework's predictions 4. Compare explanatory breadth and precision across frameworks 5. Identify complementary vs. competing predictions 6. Develop integrative synthesis where warranted

Example Comparison:

Framework	Prediction	Operationalization	Test
Resource Dependence	Mortality associated with critical resource loss	Financial Module: major client loss, funding failure	Event-mortality association

Framework	Prediction	Operationalization	Test
Population Ecology	Mortality higher in declining niches	Environment Module: market contraction	Condition-mortality association
Organizational Learning	Mortality associated with repeated errors	Dynamic Module: similar crises recurring	Pattern-mortality association

6.4 Predictive Modeling

Survival Analysis: Using Cox proportional hazards and related methods: - Time-to-failure prediction based on organizational characteristics - Risk factor identification - Subgroup survival curves

Machine Learning Approaches: Using classification and regression methods: - Mortality probability prediction from observable features - Feature importance ranking - Model validation through holdout testing

Natural Language Processing: Using computational text analysis: - Sentiment and emotion trajectories in founder narratives - Topic modeling of failure attributions - Linguistic marker analysis (certainty, agency, temporal orientation)

6.5 Qualitative Analysis

Thematic Analysis: - Coding of narrative responses for emergent themes - Pattern identification across qualitative accounts - Integration with quantitative patterns

Process Tracing: - Detailed reconstruction of mortality processes in selected cases - Mechanism identification - Counterfactual analysis

Case Comparison: - Systematic comparison of similar organizations with different outcomes - Identification of critical differentiating factors - Theory development through contrast

7. Ethical Framework

7.1 Core Principles

Dignity: Every founder and organization deserves respectful treatment. Failure is human; our methodology honors rather than exploits this reality.

Autonomy: Participants control their data. Consent is informed, specific, and revocable.

Beneficence: Research should benefit participants (through therapeutic value) and society (through knowledge generation).

Non-maleficence: Research must not harm participants, named individuals, or the broader community.

Justice: Benefits and burdens of research should be fairly distributed.

7.2 Informed Consent

Consent Elements: - Clear explanation of research purpose and methods - Description of data collected and retention - Explanation of verification process - Publication and visibility options - Rights regarding data access, correction, and deletion - Risks and benefits of participation - Voluntary participation and right to withdraw

Consent Implementation: - Written consent obtained at registration - Module-specific consent for sensitive sections - Separate consent for publication visibility tiers - Annual consent renewal for longitudinal follow-up

7.3 Privacy and Confidentiality

Data Protection: - Encryption at rest and in transit - Access controls based on role - Audit logging of all data access - Compliance with applicable data protection regulations (GDPR, CCPA, etc.)

Anonymization: - Default anonymization in all research outputs - Named publication only with explicit consent - Aggregation thresholds to prevent identification - Careful treatment of small-cell situations (unique organizations identifiable by characteristics)

Third-Party Data: - Individuals named by participants (co-founders, employees) not contacted without consent - No publication of characterizations of named individuals without their consent - Verification process respects verifier privacy

7.4 Potential Harms and Mitigations

Potential Harm	Mitigation
Emotional distress from recounting failure	Therapeutic framing, professional support available, right to pause/withdraw
Reputational damage to founders	Default anonymity, consent controls, no publication without approval
Reputational damage to named individuals	Consent required for naming, characterization review
Data misuse by third parties	No sale of individual data, aggregation requirements, ethical use commitments
False conclusions from flawed data	Transparent limitations, verification requirements, conservative claims

7.5 Therapeutic Benefit

SOIL explicitly designs for therapeutic benefit, not merely harm minimization:

Closure: The structured reflection process facilitates psychological closure on failure experience.

Meaning-Making: Narrative modules support constructive interpretation of experience.

Contribution: Framing participation as valuable contribution to knowledge provides redemption narrative.

Community: Connection with others who share similar experiences reduces isolation.

Recognition: Beautiful memorialization validates the effort invested.

Research Participants vs. Human Subjects: We view participants as co-researchers contributing knowledge, not subjects from whom data is extracted.

7.6 Research Governance

Ethics Review: - Initial protocol reviewed by independent research ethics board - Ongoing review of modifications - Adverse event reporting and response procedures

Data Access Committee: - Oversees access to individual-level data - Reviews research proposals from external researchers - Ensures ethical use commitments

Founder Advisory Council: - Includes participants in governance - Reviews methodology from participant perspective - Advises on ethical issues and community concerns

8. Limitations and Mitigation Strategies

8.1 Selection Bias

Limitation: Organizations whose founders are willing and able to participate may differ systematically from those whose founders are not. Potentially missing: founders who died, founders experiencing severe psychological distress, founders of catastrophic frauds, organizations that disappeared without identifiable founders.

Mitigation: - Multiple recruitment channels to diversify sample - Non-response analysis protocol (see Section 4.1) - Comparison with external data sources (business registries, accelerator cohort data, industry reports) - Sensitivity analysis examining how results might differ if missing cases differ systematically - Transparent reporting of sample composition and likely selection mechanisms - Propensity weighting where known selection factors can be modeled

Reporting Commitment: All analyses will include explicit discussion of how selection bias might affect conclusions, with sensitivity analyses where feasible.

8.2 Retrospective Bias

Limitation: All data is collected after organizational death, creating risks of memory decay, reconstruction, hindsight bias, and post-hoc rationalization.

Mitigation: - Temporal anchoring at salient points (Peak Operations with objective verification) - Document collection for objective verification - Multi-stakeholder triangulation - Explicit questioning about surprises and expectation violations - Separate collection of contemporaneous vs. retrospective assessments - Analysis of time-since-closure effects on reporting patterns

8.3 Self-Report Validity

Limitation: Founders may intentionally or unintentionally misrepresent their organizations.

Mitigation: - Verification requirement (3+ external confirmations) - Document verification where available - Multi-stakeholder participation for deeper cases - Pattern analysis (internally inconsistent accounts flagged) - Quality tiers with transparent reporting of verification depth

8.4 Cultural and Linguistic Variation

Limitation: Instruments developed in one cultural context may not translate equivalently across cultures. Failure meanings and attributions vary culturally.

Mitigation: - Translation and back-translation for non-English instruments - Cultural adaptation with local expert review - Regional analysis to identify cultural variation - Measurement invariance testing across cultural groups - Qualitative work to understand cultural meaning differences

8.5 Construct Drift Over Time

Limitation: Organizational forms, failure modes, and relevant functions may change over time, potentially dating instruments and findings.

Mitigation: - Regular instrument review and updating - Temporal analysis to detect changing patterns - Modular instrument design enabling updates without full redesign - Documentation of instrument versions for longitudinal tracking

8.6 Causal Inference Limitations

Limitation: Retrospective design fundamentally limits causal inference. Association $\neq$ causation.

Mitigation: - Cautious causal language in all communications - Explicit acknowledgment of alternative explanations - Pattern replication across subgroups - Temporal sequencing analysis (with appropriate caveats) - Future prospective studies to test retrospectively identified patterns

9. Research Agenda

9.1 Phase 1: Foundation (Years 1-2)

Objectives: - Build initial case database (target: 500-2,000 cases) - Validate data collection instruments - Establish baseline mortality patterns - Develop analytical infrastructure

Key Studies: - Instrument validation studies (reliability, validity) - Descriptive mortality demographics by organization type - Preliminary failure archetype identification - Verification system effectiveness assessment

Outputs: - Validated instrument battery - Baseline descriptive reports - Initial taxonomy of failure patterns - Methodological publications in research methodology journals

9.2 Phase 2: Pattern Discovery (Years 2-4)

Objectives: - Achieve statistically robust sample (target: 2,000-5,000 cases) - Identify and validate failure archetypes - Detect early warning indicators - Compare theoretical framework explanatory power

Key Studies: - Cluster analysis for failure archetype identification - Survival analysis for time-to-failure prediction - Multi-framework comparative analysis - Industry-specific mortality patterns - Geographic/cultural variation analysis

Outputs: - Failure archetype taxonomy with diagnostic criteria - Early warning indicator validation studies - Framework comparison meta-analysis - Industry-specific reports - Peer-reviewed publications in organizational science journals

9.3 Phase 3: Prediction and Prevention (Years 4-6)

Objectives: - Develop predictive models with validated accuracy - Create diagnostic assessment instruments - Test intervention protocols - Establish clinical applicability

Key Studies: - Predictive model development and validation - Diagnostic instrument development - Near-death analysis (organizations that survived similar conditions) - Intervention effectiveness studies (prospective where possible) - Prevention program development

Outputs: - Validated predictive models - Diagnostic assessment instruments - Prevention protocols - Clinical practice guidelines - Textbook on organizational mortality

9.4 Ongoing Activities

Throughout all phases:

Data Collection: - Continuous case addition - Quality improvement iterations - Instrument refinement - Sample diversification

Academic Engagement: - Conference presentations - Peer-reviewed publications - Academic partnerships - Dissertation data provision

Methodology Development: - Analytical method refinement - New instrument development - Cross-cultural validation - Temporal trend monitoring

Knowledge Translation: - Practitioner-oriented publications - Founder community education - Policy briefs where relevant - Media engagement (with ethical constraints)

10. Invitation to Collaboration

10.1 Why Collaboration is Essential

SOIL does not claim to have solved the methodological challenges of studying organizational mortality. This white paper presents our current best thinking — a foundation for critique, refinement, and collaborative development.

We recognize that: - No single team possesses all relevant expertise - Methodological blind spots are inevitable without external review - Scientific credibility requires community validation - The field will develop faster through collaboration than competition

10.2 Open Questions for Collaborative Development

We explicitly invite collaboration on the following methodological challenges:

Comparison Strategy Design > How should we optimally design survivor comparison studies? What matching criteria balance precision with feasibility? How do we address selection bias in survivor participation?

Causal Inference from Retrospective Data > What methods can strengthen causal inference from retrospective organizational data? Under what conditions can we move beyond associational claims?

Cross-Cultural Validity > How do we ensure instruments measure equivalent constructs across cultures? What cultural adaptations are necessary vs. problematic?

Longitudinal Design Integration > How can retrospective autopsy data best integrate with prospective monitoring of at-risk organizations?

AI/ML Methodological Integration > What are best practices for integrating AI-assisted data extraction and analysis while maintaining transparency and validity?

Therapeutic-Research Balance > How do we optimize the balance between therapeutic value for participants and research rigor? Where do these goals conflict, and how should conflicts be resolved?

10.3 Forms of Collaboration

Academic Partnerships - Joint research projects using SOIL data - Methodological critique and refinement - Instrument validation studies - Replication and extension

Data Access for Qualified Researchers - Anonymized dataset access for hypothesis testing - Collaborative analysis projects - Dissertation research support

Practitioner Input - Instrument validation from professional experience - Pattern validation from advisory/VC practice - Knowledge translation partnership

Methodological Review - Expert critique of instruments and protocols - Peer review of analytical approaches - Suggestions for methodological improvements

10.4 How to Engage

Researchers, practitioners, and methodologists interested in collaboration are invited to contact SOIL through:

- Academic partnership inquiries: [research@soil.rip]
- Methodological feedback: [methodology@soil.rip]
- General inquiries: [hello@soil.rip]

We commit to: - Responding to all serious inquiries - Transparent discussion of methodological limitations - Appropriate attribution in collaborative work - Open sharing of instruments and protocols for non-commercial research use

11. Conclusion

11.1 Contribution to Knowledge

SOIL addresses a fundamental gap in organizational science: the systematic, empirical study of organizational mortality. By treating organizational death with the same rigor that medicine brings to human death, we aim to build the foundation for a new discipline — Organizational Medicine.

Methodological Contributions: - Framework-agnostic data collection enabling theoretical comparison - Multi-stakeholder verification addressing self-report validity - Integration of therapeutic and research objectives - Temporal modeling distinguishing genesis, peak, and decline phases - Comprehensive coverage across organizational functions and dynamics

Substantive Contributions: - First large-scale, verified database of organizational mortality - Empirical basis for failure archetype taxonomy - Identification and validation of early warning indicators - Comparison of theoretical framework explanatory power - Foundation for predictive modeling and prevention

11.2 Practical Implications

Beyond academic contribution, SOIL aims for practical impact:

For Founders: - Therapeutic closure process - Learning from others' experiences - Reduction of failure stigma - Community and support

For Practitioners (VCs, Accelerators, Advisors): - Evidence-based portfolio risk assessment - Early warning monitoring frameworks - Intervention targeting based on mortality patterns - Due diligence enhancement

For Educators: - Empirically grounded case material - Failure literacy curriculum - Organizational health assessment training - Evidence-based entrepreneurship education

For Policymakers: - Understanding of organizational mortality in different contexts - Evidence for support program design - Ecosystem health assessment frameworks

11.3 Closing Reflection

Every organizational death represents an investment of human time, energy, creativity, and hope. Most of this investment is currently wasted — the lessons disappear with the organization. SOIL aims to transform this waste into wisdom, creating a systematic foundation for understanding why organizations die and how we might help more of them live.

The methodology presented in this white paper is not final. Science advances through iteration, critique, and improvement. We offer this framework as a serious beginning, not a definitive answer. We welcome the engagement of the scholarly community in refining these methods and building this new field together.

References

Note: This section would contain full academic citations in a published version. Key theoretical and methodological sources include:

Organizational Theory: - Hannan, M. T., & Freeman, J. (1989). Organizational Ecology - Pfeffer, J., & Salancik, G. R. (1978). The External Control of Organizations - March, J. G., & Simon, H. A. (1958). Organizations - Scott, W. R. (2013). Institutions and Organizations

Research Methodology: - Yin, R. K. (2017). Case Study Research and Applications - Creswell, J. W., & Clark, V. L. P. (2017). Designing and Conducting Mixed Methods Research - Shadish, W. R., Cook, T. D., & Campbell, D. T. (2002). Experimental and Quasi-Experimental Designs

Failure Research: - Shepherd, D. A. (2003). Learning from Business Failure - McGrath, R. G. (1999). Falling Forward: Real Options Reasoning and Entrepreneurial Failure - Ucbasaran, D., et al. (2013). Life After Business Failure

Medical Analogy: - Virchow, R. (1858). Cellular Pathology - Foucault, M. (1963). The Birth of the Clinic

Appendices

Appendix A: Complete Function Taxonomy

[Full listing of all functions by organization type, with definitions and measurement specifications]

Appendix B: Event Category Definitions

[Complete taxonomy of internal, external, financial, and personal event categories with coding instructions]

Appendix C: Sample Instruments

[Example modules showing exact question wording and response options]

Appendix D: Verification Protocol

[Detailed verification procedures including verifier communication templates]

Appendix E: Analytical Codebooks

[Coding schemes for qualitative data analysis]

Appendix F: Ethics Documentation

[Informed consent templates, ethics board approvals, data protection policies]

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“Just as the systematic study of human death gave rise to modern medicine, so the systematic study of organizational death may give rise to organizational medicine — the scientific understanding, prediction, and prevention of organizational mortality.”